MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

PHYS-111: Lecture 2 Tutorial questions.

1. A particle moves according to the equation $x = 10t^2$ where x is in meters and t is in seconds.
   (a) Find the average velocity for the time interval from 2.00 s to 3.00 s.
   (b) Find the average velocity for the time interval from 2.00 s to 2.10 s.
2. The position of a pinewood derby car was observed at various times; the results are summarized in the following table. Find the average velocity of the car for
   (a) The first second
   (b) The last 3 s,
   (c) The entire period of observation.

t(s)	0	1.0	2.0	3.0	4.0	5.0
x(m)	0	2.3	9.2	20.7	36.8	57.5

3. Find the instantaneous velocity of the particle described in Figure 1 at the following times:
   (a) $t = 1.0\ s$, (b) $t = 3.0\ s$, (c) $t = 4.5\ s$, and (d) $t = 7.5\ s$.

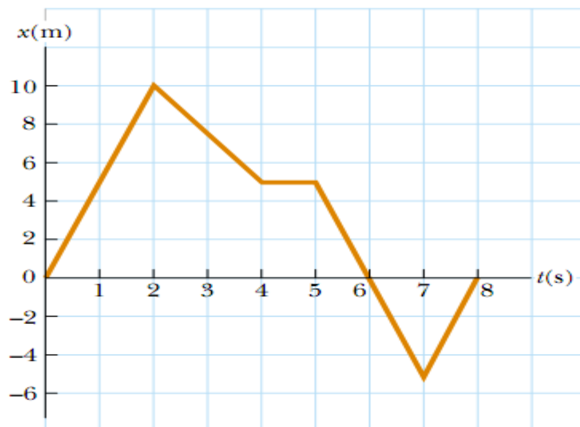

Figure 1

4. A velocity–time graph for an object moving along the x-axis is shown in Figure 2.
   (a) Plot a graph of the acceleration versus time.
   (b) Determine the average acceleration of the object in the time intervals $t = 5.00\ s$ to $t = 15.0\ s$ and $t = 0$ to $t = 20.0\ s$.

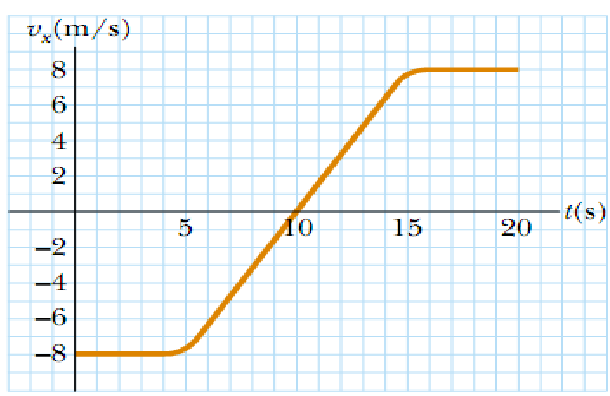

Figure 2

5. A truck covers 40.0 m in 8.50 s while smoothly slowing down to a final speed of 2.80 m/s.
   (a) ncFind its original speed.
   (b) Find its acceleration.
6. Starting from a pillar, you run 200 m east (the +x-direction) at an average speed of 5.0 m/s and then run 280 m west at an average speed of 4.0 m/s to a post. Calculate:
   (a) Your average speed from pillar to post
   (b) Your average velocity from pillar to post.
7. An electron in a cathode ray tube (CRT) accelerates from $2.00 \times 10^4\ m/s$ to $6.00 \times 10^6\ m/s$ over 1.50 cm.
   (a) How long does the electron take to travel this 1.50 cm?
   (b) What is its acceleration?
8. In the fastest measured tennis serve, the ball left the racquet at 73.14 m/s. A served tennis ball is typically in contact with the racquet for 30.0 ms and starts from rest. Assume constant acceleration.
   (a) What was the ball's acceleration during this serve?
   (b) How far did the ball travel during the serve?
9. In Mangochi, Malawi, the ultimate test of a young man's courage once was to jump off a bridge into the Shire River, 23.0 m below the bridge.
   (a) How long did the jump last?
   (b) How fast was the diver traveling upon impact with the water?
   (c) If the speed of sound in air is 340 m/s, how long after the diver took off did a spectator on the bridge hear the splash?
10. A rock is thrown straight up and reaches a height of 10 m.
   (a) How long was the rock in the air?
   (b) What is the initial velocity of the rock?
11. A student throws a set of keys vertically upward to her sorority sister, who is in a window 4.00 m above. The second student catches the keys 1.50 s later
   (a) With what initial velocity were the keys thrown?
   (b) What was the velocity of the keys just before they were caught?
12. A basketball dropped from rest 1.00 m above the floor rebounds to a height of 0.67 m. Assuming the ball is not moving horizontally, calculate its velocity…
   (a) just before it hits the floor on the way down and
   (b) Just after it left the floor on the way up.
   (c) If the ball is in contact with the floor for 0.10 s determine its acceleration…
      i. On the way down,
      ii. While it is contact with the floor, and
      iii. On the way up.